

DERIVATIVES PRICING

A COMPUTATIONAL & THEORETICAL FRAMEWORK

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Part I

Mathematical Foundations

1 Measure and Integration

2 Probability

3 Convergence

4 Conditioning

5 Martingales

6 Wiener and Brownian Motion Processes

7 Itô Calculus

7.1 Itô Integral

We will work with a Wiener process W on a complete probability space $(\Omega, \mathcal{H}, \mathbb{P})$ with a right-continuous filtration $\mathcal{F} := (\mathcal{F}_s, s \geq 0)$ and such that W is adapted to $\mathcal{F}$. We also use the following definition.

Definition 7.1: Simple Processes

A process $F := (F_s, s \in [0, t])$ is called *simple* if it is of the form

$$F_s := \xi_0 \mathbb{1}_{\{0\}}(s) + \sum_{k=0}^{n-1} \xi_k \mathbb{1}_{(s_k, s_{k+1}]}(s), \quad s \in [0, t], \quad (7.1)$$

where $\{s_0, \dots, s_n : 0 = s_0 < \dots < s_n = t\} =: \Pi_n, n \in \mathbb{N}$, is a segmentation of $[0, t]$ and $\xi_k \in \mathcal{F}_{s_k}$ for all k .

Note that the simple process in Equation (7.1) is right-continuous at $s = 0$ and is left-continuous at $s \in (0, t]$. Figure 1 shows a realisation of a simple process.

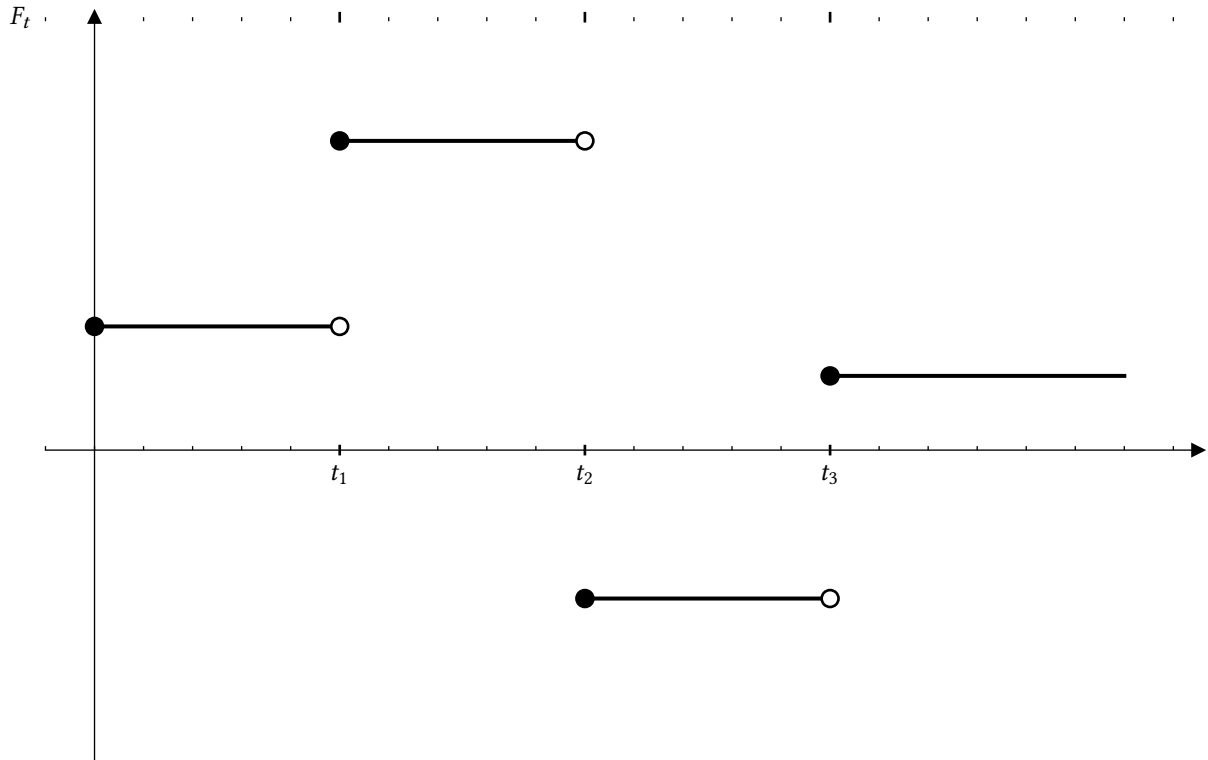


Figure 1: A path of a simple process.

We will consider the interplay between the simple process F_t and the Brownian motion $W(t)$ in the following way. Regard $W(t)$ as the price per share of an asset at time t . Since Brownian motion can take both negative and positive values, it is not a suitable model for the price of a limited-liability asset, such as a stock. Notwithstanding, for the sake of this illustration, we ignore that issue. Think of $t_0, t_1, \dots, t_{n-1}$ as the *trading dates* in the asset, and think of $F_{t_0}, F_{t_1}, \dots, F_{t_{n-1}}$ as the position (number of shares) taken in the asset at each trading date and held to the next trading date. The gain from trading at each time t is given by

$$I(t) = F_{t_0} [W(t) - W(t_0)] = F_0 W(t), \quad 0 \leq t \leq t_1, \quad (7.2)$$

$$I(t) = F_0 W(t_1) + F_{t_1} [W(t) - W(t_1)], \quad t_1 \leq t \leq t_2, \quad (7.3)$$

$$I(t) = F_0 W(t_1) + F_{t_1} [W(t_2) - W(t_1)] + F_{t_2} [W(t) - W(t_2)], \quad t_2 \leq t \leq t_3. \quad (7.4)$$

and so on. In general, if $t_k \leq t \leq t_{k+1}$, then

$$I(t) = \sum_{j=0}^{k-1} F_{t_j} [W(t_{j+1}) - W(t_j)] + F_{t_k} [W(t) - W(t_k)]. \quad (7.5)$$

We are now ready to define the Itô Integral for simple processes

Definition 7.2: Itô Integral for Simple Processes

Suppose that F is a simple process of the form Equation (7.1), with $\int_0^t \mathbb{E} F_s^2 ds < \infty$.

Then, the *Itô integral* of F with respect to W over $[0, r]$ is defined as

$$\int_0^r F_s dW_s := \sum_{k=0}^{n-1} \xi_k (W_{s_{k+1} \wedge r} - W_{s_k \wedge r}), \quad r \in [0, t]. \quad (7.6)$$

The Itô integral from r to t is naturally defined as the difference of two integrals:

$$\int_r^t F_s dW_s = \int_0^t F_s dW_s - \int_0^r F_s dW_s. \quad (7.7)$$

The Itô integral has the important *isometry* property:

$$\mathbb{E} \left[\left(\int_0^t F_s dW_s \right)^2 \right] = \int_0^t \mathbb{E} [F_s^2] ds, \quad (7.8)$$

which allows us to compute the variance of the Itô integral. In fact, the Itô integral satisfies the following four important properties:

Theorem 7.1: Properties of the Itô Integral

Assume that the Itô integral is well-defined. Then, for $\alpha, \beta \in \mathbb{R}$ and $r \in [0, t]$:

- (1) (Linearity): $\int_0^r [\alpha F_s + \beta G_s] dW_s = \alpha \int_0^r F_s dW_s + \beta \int_0^r G_s dW_s$.
- (2) (Zero mean): $\mathbb{E} \int_0^r F_s dW_s = 0$.
- (3) (Isometry): $\mathbb{E} \left[\int_0^r F_s dW_s \times \int_0^r G_s dW_s \right] = \int_0^r \mathbb{E} [F_s G_s] ds$.
- (4) (Martingale): $(\int_0^r F_s dW_s, r \in [0, t])$ is an L^2 -martingale.

Definition 7.3: Canonical Approximation

For any $\mathcal{F}$ -adapted process $F := (F_s, s \in [0, t])$, we define its *canonical approximation* as

$$F_s^{(n)} := F_0 \mathbb{1}_{\{0\}}(s) + \sum_{k=0}^{n-1} F_{s_k} \mathbb{1}_{(s_k, s_{k+1}]}(s), \quad s \in [0, t]. \quad (7.9)$$

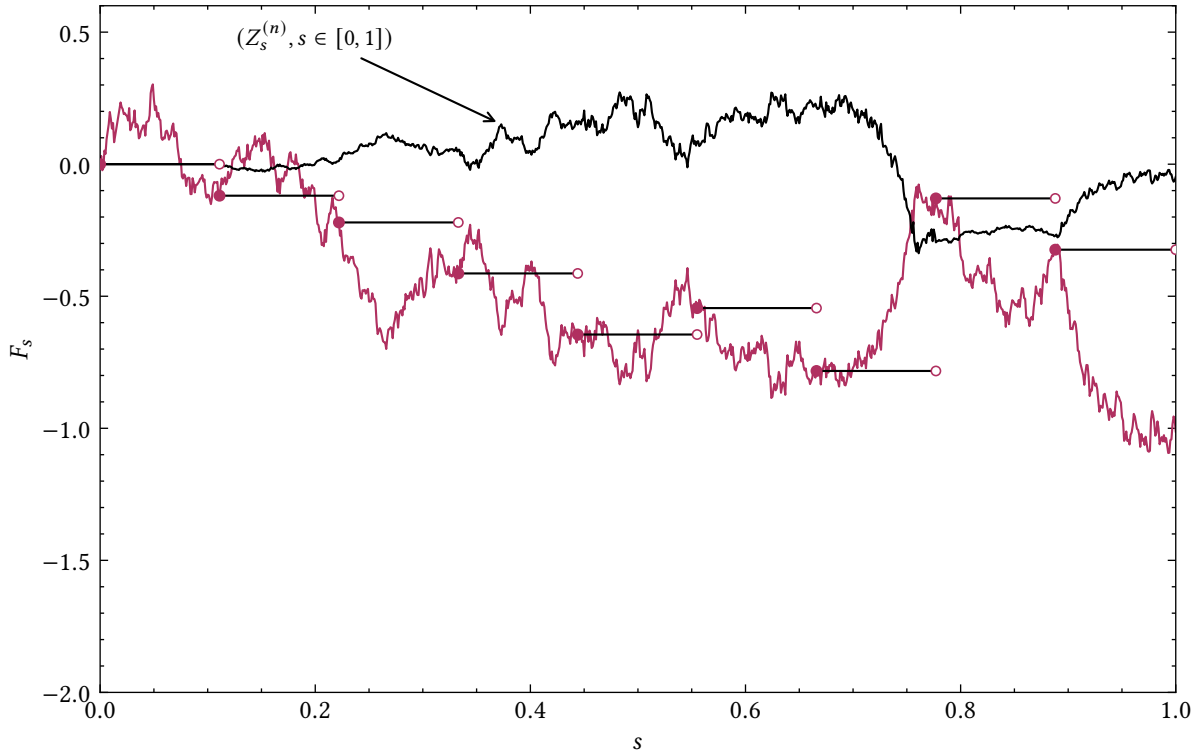


Figure 2: A Wiener process $F := (W_s, s \in [0, t])$, its canonical approximation $F^{(n)}$ and the Itô integral process $Z^{(n)} = (\int_0^r F_s^{(n)} dW_s, r \in [0, 1])$

Having defined the Itô integral for simple processes, we are now ready to extend the definition to more general predictable processes.

7.1.1 Itô Integral for Predictable Processes

While simple processes of the form Equation (7.1) are $\mathcal{F}$ -predictable, it is not immediately clear if more general processes are also $\mathcal{F}$ -predictable. The following result shows that any process $F := (F_s, s \in [0, t])$ adapted to $\mathcal{F}$ and left-continuous on $(0, t]$ is not only $\mathcal{F}$ -predictable, but also *indistinguishable* from any other left-continuous modification $\tilde{F}$ of F ; that is, if $\mathbb{P}(F_s = \tilde{F}_s) = 1$ for all $s \in [0, t]$, then $\mathbb{P}(F_s = \tilde{F}_s \text{ for all } s \in [0, t]) = 1$. Moreover,

F is *progressively measurable*; that is, the mapping $F : [0, t] \times \Omega \rightarrow \mathbb{R}$ is measurable with respect to the product σ -algebra $\mathcal{B}_{[0,t]} \otimes \mathcal{F}_t$.

Theorem 7.2: Properties of Left-continuous Adapted Processes

Suppose that $F := (F_s, s \in [0, t])$ is a process on $(\Omega, \mathcal{H}, \mathbb{P})$ that is adapted to the filtration $\mathcal{F}$ and is left-continuous on $(0, t]$. Then, the following hold:

- (1) (Predictable): The process F is $\mathcal{F}$ -predictable (i.e., $\mathcal{F}^P$ -measurable).
- (2) (Progressively Measurable): For each t , the mapping $F : [0, t] \times \Omega \rightarrow \mathbb{R}$ is jointly measurable with respect to $\mathcal{B}_{[0,t]} \otimes \mathcal{F}_t$.
- (3) (Indistinguishable): If $\tilde{F}$ is a modification of F and almost surely left-continuous, then $\mathbb{P}(\cap_{s \leq t} \{F_s = \tilde{F}_s\}) = 1$.

Proof. Consider the canonical approximation $F^{(n)}$ of F , given in (7.9), where the segmentation Π_n satisfies $\text{mesh}(\Pi_n) \rightarrow 0$.

For $s = 0$ and $\omega \in \Omega$, we have $F_0(\omega) = F_0^{(n)}(\omega)$ by construction. For any $s \in (0, t]$, choose k (depending on n and s) with $s_k < s \leq s_{k+1}$. Define

$$\varepsilon_{n,s} := s - s_k, \quad 0 < \varepsilon_{n,s} \leq \text{mesh}(\Pi_n). \quad (7.10)$$

Then

$$F_s - F_s^{(n)} = F_s - F_{s_k} + \underbrace{F_{s_k} - F_s^{(n)}}_{=0} = F_s - F_{s-\varepsilon_{n,s}}, \quad (7.11)$$

and the last difference vanishes as $n \uparrow \infty$ because F is left-continuous. Hence, $F_s^{(n)}(\omega) \rightarrow F_s(\omega)$ for all $s \in [0, t]$ and $\omega \in \Omega$.

Since F is adapted to $\mathcal{F}$, Example 7.4 implies that each simple process $F^{(n)}$ is $\mathcal{F}$ -predictable. Consequently, for each t , the mapping $F^{(n)} : [0, t] \times \Omega \rightarrow \mathbb{R}$ is jointly measurable with respect to $\mathcal{B}_{[0,t]} \otimes \mathcal{F}_t$. Because pointwise limits of measurable functions are measurable, the same properties hold for the limit F . This establishes $\mathcal{F}$ -predictability and progressive measurability for F .

Now assume that $\widetilde{F}$ is a modification of F ; that is, $\mathbb{P}(\widetilde{F}_s \neq F_s) = 0$ for all $s \in [0, t]$. Let

$$B := \{\text{both } F \text{ and } \widetilde{F} \text{ are left-continuous on } (0, t]\}, \quad (7.12)$$

so that $\mathbb{P}(B) = 1$. Let $\mathbb{Q}_t := [0, t] \cap \mathbb{Q}$ and define

$$A := \bigcap_{s \in \mathbb{Q}_t} \{\widetilde{F}_s = F_s\}. \quad (7.13)$$

Since $\mathbb{Q}_t$ is countable, by countable subadditivity ??,

$$\mathbb{P}(A^c) = \mathbb{P}\left(\bigcup_{s \in \mathbb{Q}_t} \{\widetilde{F}_s \neq F_s\}\right) \leq \sum_{s \in \mathbb{Q}_t} \mathbb{P}(\widetilde{F}_s \neq F_s) = 0, \quad (7.14)$$

so $\mathbb{P}(A) = 1$.

By denseness of $\mathbb{Q}_t$, for any $s \in [0, t]$ there exists a sequence $(s_n) \subset \mathbb{Q}_t$ with $s_n \uparrow s$.

Therefore, for all $\omega \in A \cap B$,

$$F_s(\omega) - \widetilde{F}_s(\omega) = \lim_{n \uparrow \infty} (F_{s_n}(\omega) - \widetilde{F}_{s_n}(\omega)) = 0, \quad \forall s \in [0, t]. \quad (7.15)$$

Since $\mathbb{P}(A \cap B) = 1$, we conclude that F and $\widetilde{F}$ are indistinguishable. $\square$

We have so far considered stochastic processes that are left-continuous and adapted to the filtration $\mathcal{F}$. However, in order to define the Itô integral for these more general processes, additional regularity conditions are needed.

Definition 7.4: The Class of Processes $\mathcal{P}_t$

Let $\mathcal{P}_t$ be the class of stochastic processes $F := (F_s, s \in [0, t])$ that are left-continuous on $(0, t]$, $\mathcal{F}$ -adapted, and satisfy the norm condition:

$$\|F\|_{\mathcal{P}_t}^2 := \mathbb{E} \int_0^t F_s^2 ds < \infty.$$

Theorem 7.3: Approximating the Class $\mathcal{P}_t$ via Simple Processes

Suppose that $F \in \mathcal{P}_t$. Then, there exists a sequence $(F^{(n)})$ of simple processes of the form Equation (7.1) such that $\text{mesh}(\Pi_n) \rightarrow 0$ implies that $F_s^{(n)} \xrightarrow{\text{a.s.}} F_s$ for all $s \in [0, t]$ and

$$\lim_{n \rightarrow \infty} \|F^{(n)} - F\|_{\mathcal{P}_t} = 0. \quad (7.16)$$

Proposition 7.1: Deterministic Integrands and Gaussian Processes

Let $f : \mathbb{R}_+ \times \mathbb{R}_+ \rightarrow \mathbb{R}$ be left-continuous in the first argument and such that $\|f(\cdot, t)\|_{\mathcal{P}_t}^2 = \int_0^t f^2(s, t) ds < \infty$. Then, the stochastic integral process

$$Z_t := \int_0^t f(s, t) dW_s, \quad t \geq 0, \quad (7.17)$$

is a zero-mean Gaussian process with covariance function

$$\text{Cov}(Z_{t_1}, Z_{t_2}) = \int_0^{t_1 \wedge t_2} f(s, t_1) f(s, t_2) ds, \quad t_1, t_2 \geq 0. \quad (7.18)$$

$$\lim_{m \rightarrow \infty} \sup_{s \in [0, t]} \mathbb{E} F_s^2 \mathbf{1}_{\{F_s^2 > m\}} = 0. \quad (7.19)$$

Lemma 7.1: Quadratic Variation Integral

Suppose that $F \in \mathcal{P}_t$ satisfies the condition Equation (7.19). Then, as $\text{mesh}(\Pi_n) \rightarrow 0$:

$$\sum_{k=0}^{n-1} F_{s_k} (W_{s_{k+1}} - W_{s_k})^2 \xrightarrow{L^2} \int_0^t F_s ds. \quad (7.20)$$

Theorem 7.4: Enlarging the Class $\mathcal{P}_t$ via Localization

Let $F := (F_s, s \in [0, t])$ be an $\mathcal{F}$ -adapted process, left-continuous on $(0, t]$, and such that (7.34) holds. Then, there exists a localizing sequence (T_n) such that the stopped process $(\int_0^{r \wedge T_n} F_s dW_s, r \in [0, t])$ is a square-integrable martingale. In addition, the Itô integral of F with respect to W on $[0, t]$ is defined as the almost sure limit:

$$\int_0^t F_s dW_s := \lim_{n \rightarrow \infty} \int_0^{t \wedge T_n} F_s dW_s. \quad (7.21)$$

Proposition 7.2: Continuous Modification and Stopping Times

If $F \in \mathcal{P}_t$ and $T \leq t$ is a stopping time with respect to $\mathcal{F}$, then almost surely:

$$\int_0^T F_s dW_s = \int_0^t \mathbb{1}_{[0, T]}(s) F_s dW_s. \quad (7.22)$$

Theorem 7.5: Continuous Modification of the Itô Integral

Suppose that $F \in \mathcal{P}_t$ and $(F^{(n)})$ is a sequence of simple processes satisfying (7.17). Denote the limit $\int_0^r F_s^{(n)} dW_s =: Z_r^{(n)} \xrightarrow{L^2} Z_r$ as the Itô integral on $[0, r]$ for $r \leq t$. Then, there exists a modification $\tilde{Z}$ of $(Z_r, r \in [0, t])$ such that the mapping $r \mapsto \tilde{Z}_r$ is almost surely continuous. Moreover, this modification is unique up to indistinguishability.

Theorem 7.6: Existence of the Itô Integral with Integrand $F \in \mathcal{P}_t$

Suppose that $F \in \mathcal{P}_t$ and $(F^{(n)})$ is a sequence of simple processes satisfying (7.17). Then, the following limit exists in L^2 norm:

$$\int_0^t F_s dW_s := \lim_{n \rightarrow \infty} \int_0^t F_s^{(n)} dW_s, \quad r \in [0, t], \quad (7.23)$$

and it defines the Itô integral of F with respect to W on $[0, t]$.

7.2 Itô Calculus

Theorem 7.7: Itô Formula for the Wiener Process

Let W be a Wiener process and $f : \mathbb{R} \rightarrow \mathbb{R}$ be twice continuously differentiable with derivatives f' and f'' . Then, almost surely

$$f(W_t) - f(W_0) = \int_0^t f'(W_s) dW_s + \frac{1}{2} \int_0^t f''(W_s) ds.$$

Definition 7.5: Integral With Respect To An Itô Process

Let X be an $\text{Itô}(\mu, \sigma)$ process and F a left-continuous $\mathcal{F}$ -adapted process such that almost surely $\int_0^t (|F_s \mu_s| + (F_s \sigma_s)^2) ds < \infty$ for all t . Then, the integral of F with respect to X is defined as the $\text{Itô}(F\mu, F\sigma)$ process with:

$$\int_0^t F_s dX_s := \int_0^t F_s \mu_s ds + \int_0^t F_s \sigma_s dW_s. \quad (7.24)$$

Definition 7.6: Quadratic Variation On $[0, t]$

Let $\Pi_n := \{s_k, k = 0, \dots, n\}$ be a segmentation of $[0, t]$ and let $(X^{(n)})$ be a sequence of simple processes with almost sure limit $X_s := \lim_n X_s^{(n)}$ for all $s \in [0, t]$ as $\text{mesh}(\Pi_n) \rightarrow 0$. The *quadratic variation* of $X := (X_s, s \in [0, t])$ over $[0, t]$, denoted as $\langle X \rangle_t$, is defined as the limit in probability of

$$\langle X^{(n)} \rangle_t := \sum_{k=0}^{n-1} (X_{s_{k+1}}^{(n)} - X_{s_k}^{(n)})^2. \quad (7.25)$$

Theorem 7.8: Quadratic Variation Of An Itô Process

If X is an Itô(μ, σ) process, then

$$\langle X^{(n)} \rangle_t \xrightarrow{L^2} \langle X \rangle_t = \int_0^t \sigma_s^2 ds. \quad (7.26)$$

Furthermore, if $F \in \mathcal{P}_t$ satisfies the condition (7.19) and $\text{mesh}(\Pi_n) \rightarrow 0$, then:

$$\sum_{k=0}^{n-1} F_{s_k} (X_{s_{k+1}} - X_{s_k})^2 \xrightarrow{L^2} \int_0^t F_s \sigma_s^2 ds. \quad (7.27)$$

Theorem 7.9: Itô's Formula For Itô Processes

Let X be an Itô process and let f be twice continuously differentiable. Then,

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) d\langle X \rangle_s. \quad (7.28)$$

Definition 7.7: Covariation Of Two Itô Processes

Let X and Y be two Itô processes adapted to the same filtration $\mathcal{F}$. Then, the *covariation* between X and Y is defined as the quantity:

$$\langle X, Y \rangle_t := \frac{\langle X + Y \rangle_t - \langle X \rangle_t - \langle Y \rangle_t}{2}. \quad (7.29)$$

Theorem 7.10: Itô's Formula In $\mathbb{R}^d$

Let (X_t) be a d -dimensional Itô process, and let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be twice continuously differentiable in all variables. Then,

$$df(X_t) = \sum_{i=1}^d \partial_i f(X_t) dX_{t,i} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d \partial_{ij} f(X_t) d\langle X_{\cdot,i}, X_{\cdot,j} \rangle_t. \quad (7.30)$$

Theorem 7.11: Itô Processes And Martingales

An Itô process in $\mathbb{R}^d$ is a local martingale if and only if its drift is 0.

7.3 Stochastic Differential Equations

Stochastic differential equations are based on the same principle as ordinary differential equations, relating an unknown function to its derivatives, but with the additional feature that part of the unknown function is driven by randomness.

Definition 7.8: Stochastic Differential Equation

A *stochastic differential equation* (SDE) for a stochastic process $(X_t, t \geq 0)$ is an expression of the form

$$dX_t = a(X_t, t)dt + b(X_t, t)dW_t, \quad (7.31)$$

where $(W_t, t \geq 0)$ is a Wiener process and a and b are deterministic functions, called the *drift* and *diffusion* functions. The resulting process $(X_t, t \geq 0)$ is called an *(Itô) diffusion*.

Consider the autonomous SDE

$$dX_t = a(X_t)dt + b(X_t)dW_t. \quad (7.32)$$

As already mentioned, the meaning of this SDE is that X is a stochastic process that satisfies the integral equation

$$X_t = X_0 + \int_0^t a(X_s)ds + \int_0^t b(X_s)dW_s. \quad (7.33)$$

However, it is not clear that such a process indeed exists. We show in this section that under mild conditions on a and b we can obtain a unique solution X of the SDE via a method of successive approximations, similar to *Picard iteration* for ordinary differential equations.

The condition that is placed on a and b is that they be *Lipschitz continuous*; that is, for all $x \in \mathbb{R}$ there exists a constant c such that

$$|a(y) - a(x)| + |b(y) - b(x)| \leq c|y - x|. \quad (7.34)$$

The successive approximation idea is very simple. Suppose that $X_0 = x$. Define $X_t^{(0)} := x$ for all t , and let

$$X_t^{(n+1)} := x + \int_0^t a(X_s^{(n)}) ds + \int_0^t b(X_s^{(n)}) dW_s \quad (7.35)$$

for $n \in \mathbb{N}$. The following is the main existence result for SDEs:

Theorem 7.12: Existence of SDE Solutions

For any $t \in [0, v]$, the sequence $(X^{(n)})$ in (7.35) converges almost surely in the uniform norm on $[0, t]$ to a process X that satisfies the integral equation (7.33).

Lemma 7.2: Integral Bound

Let $X^{(n)}, n = 1, 2, \dots$ be defined in (7.35), with $X^{(0)} = x$. Then, for every $v > 0$ there exists a constant α such that

$$\mathbb{E} \sup_{s \leq t} \left(X_s^{(n+1)} - X_s^{(n)} \right)^2 \leq \alpha \|X^{(n)} - X^{(n-1)}\|_{\mathcal{P}_t}^2, \quad t \leq v.$$

Definition 7.9: Itô Stratonovich Integral

Let X and Y be Itô processes such that $\int_0^t Y_s dX_s$ and $\int_0^t X_s dY_s$ are well-defined stochastic integrals. Then, the *Itô Stratonovich integral* of Y with respect to X is defined as

$$\int_0^t Y_s \circ dX_s := \int_0^t Y_s dX_s + \frac{1}{2} \langle X, Y \rangle_t. \quad (7.36)$$

Proposition 7.3: Itô Stratonovich Formula

Let (X_t) be a d -dimensional Itô process, and let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be three times continuously differentiable in all variables. Then,

$$df(X_t) = \sum_{i=1}^d \partial_i f(X_t) \circ dX_{t,i}. \quad (7.37)$$

$$\widetilde{\mathbb{P}}^t(A) := \mathbb{E} M_t \mathbf{1}_A, \quad A \in \mathcal{F}_t. \quad (7.38)$$

$$M_t = \exp \left(\int_0^t \mu_s dW_s - \frac{1}{2} \int_0^t \mu_s^2 ds \right) = e^{Z_t - \frac{1}{2} \langle Z \rangle_t}, \quad t \geq 0, \quad (7.39)$$

$$\lim_{\varepsilon \downarrow 0} \varepsilon \ln \mathbb{E} \exp((1 - \varepsilon)\langle Z \rangle_t / 2) = 0. \quad (7.40)$$

7.3.1 Methods for Solving Simple SDEs

If a strong solution of the Itô diffusion SDE exists, then it can be expressed as $X_t = \phi(X_0, (W_s, s \in [0, t]), t)$ for some function ϕ . Finding an analytical solution for ϕ is usually impossible, however, except in a few special cases.

7.3.1.1 Linear Stochastic Differential Equations One setting where the strong solution of an SDE can be expressed as a simple analytical formula is when both the drift and diffusion of the SDE are linear functions of the state variables. In particular, suppose that

$$a(x, t) = \alpha_t x + \beta_t,$$

$$b(x, t) = \delta_t x + \gamma_t.$$

An SDE of the form

$$dX_t = (\alpha_t + \beta_t X_t)dt + (\delta_t + \gamma_t X_t)dW_t \quad (7.41)$$

is said to be *linear*.

The solution to the general linear SDE in Equation (7.41) is given next.

Proposition 7.4: Solution to a Linear Diffusion SDE

The solution to the linear SDE Equation (7.41) is given by

$$X_t = Y_t \left(X_0 + \int_0^t \frac{\alpha_s - \delta_s \gamma_s}{Y_s} ds + \int_0^t \frac{\delta_s}{Y_s} dW_s \right). \quad (7.42)$$

Where

$$Y_t = \exp \left(\int_0^t \left(\beta_s - \frac{\gamma_s^2}{2} \right) ds + \int_0^t \gamma_s dW_s \right) \quad (7.43)$$

is the stochastic exponential of an Itô(β, γ) process.

Theorem 7.13: Girsanov's Change of Measure

Suppose that μ satisfies the Krylov condition [Equation \(7.40\)](#) and M is the exponential martingale [Equation \(7.39\)](#). Then, under the change of measure [Equation \(7.38\)](#) induced by M , the Itô process

$$\tilde{W}_t := - \int_0^t \mu_s ds + W_t, \quad t \geq 0$$

is a Wiener process.

A fundamental stochastic process in finance is defined as the solution to the SDE

GBM.py

```

1  # Generate Brownian Motion
2  t = np.linspace(0, T, N + 1)
3  W = np.concatenate(([0], np.cumsum(dW)))
4
5  # Exact Solution
6  S_t = S_0 * np.exp((mu - 0.5 * sigma**2) * t + sigma * W)

```

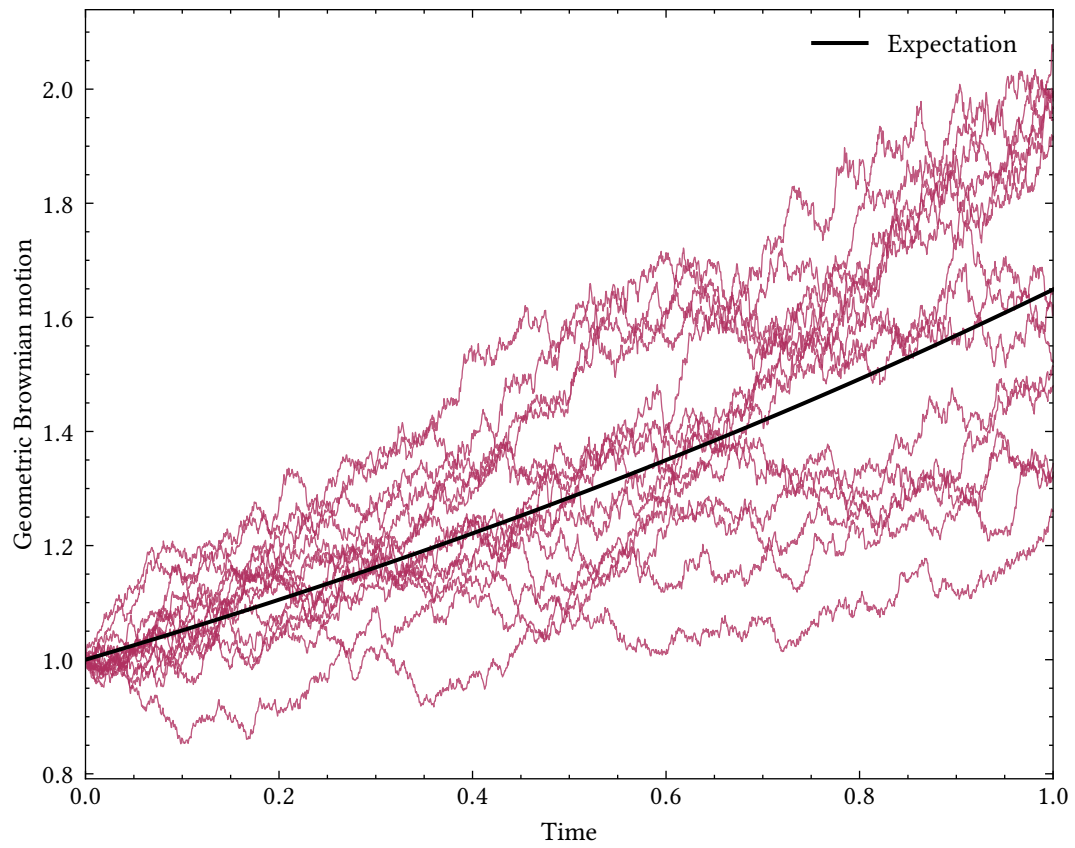


Figure 3: Geometric Brownian motion sample path and expectation for $\mu = 0.5$, $\sigma = 0.2$ and $S_0 = 1$.

Part II

Levy Processes

Part III

Stochastic Volatility

Part IV

Risk-Neutral Valuation

8 Discrete-time Market Model

9 Continuous-time Market Model

Part V

Martingale Methods

10 The Black-Scholes Model

11 A Dividend-paying Stock

12 Discretisation Schemes

13 Connection to Partial Differential Equations

Part VI

Monte Carlo Methods

SIMULATION techniques are invaluable in managing complex financial scenarios that involve path-dependent payoffs or distributions without analytical expressions. One prominent application is pricing derivative securities, estimating value at risk, and emulating hedging strategies. Monte Carlo simulation, in particular, proves highly effective for such tasks, offering a blend of simplicity and adaptability as its primary advantages. However, the method's drawback lies in its considerable computational demands, which can be mitigated to some extent. Our approach employs Monte Carlo simulation to generate simulated option price values, which are then compared against outcomes derived from analytical formulas.

14 Generating Random Numbers and Random Variables

15 Quasi Monte Carlo

IT is customary in this setting to focus on the problem of numerical integration over the unit hypercube.

The digital constructions are similar to feedback shift register random number generators, while lattice rules are similar to congruential generators. One useful way to think of QMC is that we are utilising a small random number generator in its entirety. The tools used to develop and analyse low-discrepancy methods are very different from those used in ordinary Monte Carlo, as they draw on number theory and abstract algebra rather than probability and statistics.

15.1 Discrepancy Measures

15.2 Low-discrepancy Sequences

15.2.1 Halton and Hammersley

15.2.2 Faure

15.2.3 Sobol

15.3 Koksma-Hlawka

16 Variance Reduction

Part VII

Grid & Lattice Methods

17 Lattice Methods

18 Finite Difference Methods

19 Contract Sensitivities

19.1 Likelihood Ratio

19.2 Pathwise Derivatives

19.3 Finite-Difference Approximations

Part VIII

Financial Derivatives

20 European Options

21 Exotic Options

22 American Options

European option contracts specify an expiration date, and if the option is to be exercised at all, the exercise must occur on the expiration date. An option whose owner can choose to exercise at any time up to and including the expiration date is called *American*.

Whereas a European option can be exercised only at a fixed date, an American option can be exercised any time up to its expiration. The value of an American option is the value achieved by exercising optimally. Finding this value entails finding the optimal exercise rule by solving an optimal stopping problem and computing the expected discounted payoff of the option under this rule. The embedded optimisation problem makes this a difficult problem for simulation.

Because an American option can be exercised at any time prior to its expiration, it can never be worth less than the payoff associated with immediate exercise. This is called the *intrinsic value* of the option.

To price an American option, just as with a European option, we could imagine selling the option in exchange for some initial capital and then consider how to use this capital to hedge the short position in the option. In this case, we would need to be ready to pay off the option at all times prior to the expiration date because we do not know when it will be exercised. We could determine when, from our point of view, is the worst time for the owner to exercise the option. From the owner's point of view, this would be the *optimal exercise time*, and we shall call it that. We could then compute the initial capital we need in order to be hedged against exercise at the optimal exercise time. Finally, we could show how to invest this capital so that we are hedged even if the owner exercises at a nonoptimal time. In the subsequent sections, we do all these things but begin the analysis at a different point than for European options. We define the price of American options using a risk-neutral pricing formula and then show that this price is the smallest initial capital that permits construction of the hedge just described.

22.1 Perpetual American Put

The simplest interesting American option is the *perpetual American put*. It is interesting because the optimal exercise policy is not obvious, and it is simple because this policy can be determined explicitly. Although this is not a traded option, we begin our discussion with it in order to present in a simple context the ideas behind the subsequent analysis of more realistic options. The underlying asset in most of this chapter (except where the asset pays dividends, which will be explicitly mentioned) has the price process $S(t)$ given by

$$dS(t) = rS(t), dt + \sigma S(t), d\tilde{W}(t), \quad (8.3.1)$$

where the interest rate r and the volatility σ are strictly positive constants and $\tilde{W}(t)$ is a Brownian motion under the risk-neutral probability measure $\tilde{\mathbb{P}}$. The perpetual American put pays $K - S(t)$ if it is exercised at time t . This is its intrinsic value.

Definition 22.1: Perpetual American Put

Let $\mathcal{T}$ be the set of all stopping times. The price of the perpetual American put is defined to be

$$v_*(x) = \max_{\tau \in \mathcal{T}} \widetilde{\mathbb{E}} [e^{-r\tau} (K - S(\tau))], \quad (22.1)$$

where $x = S(0)$ in (8.3.2) is the initial stock price. In the event that $\tau = \infty$, we interpret $e^{-r\tau} (K - S(\tau))$ to be zero.

The idea behind [Definition 22.1](#) is that the owner of the perpetual American put can choose an exercise time τ , subject only to the condition that she may not look ahead to determine when to exercise. The mathematical formulation of this “not look ahead” restriction is that τ must be a stopping time. The price of the option at time zero is the risk-neutral expected payoff of the option, discounted from the exercise time back to time zero. If the option is never exercised, its payoff is zero. This explains the term under the expectation on the right-hand side of [Equation \(22.1\)](#). The owner of the option should choose the exercise strategy that maximizes this expected payoff, discounted back to time zero, and thus we define the price of the option to be the maximum over $\tau \in \mathcal{T}$ of the discounted expected payoffs.

The owner of the perpetual American put can exercise at any time. In particular, there is no expiration date after which the put can no longer be exercised. This makes every date like every other date; the time remaining to expiration is always the same (i.e., infinity). Because every date is like every other date, it is reasonable to expect that the optimal exercise policy depends only on the value of $S(t)$ and not on the time variable t . The owner of the put should exercise as soon as $S(t)$ falls far enough below K . In other words, it is reasonable to expect that the optimal exercise policy is of the form

Exercise the put as soon as $S(t)$ falls to the level L_* .

We have two questions to answer:

- (i) What is the value of L_* , and how do we know it corresponds to optimal exercise?
- (ii) What is the value of the put?

For the perpetual American put, we can base the answers to these questions on explicit computations.

22.2 Approximate Continuation Values

Regression-based methods posit an expression for the continuation value of the form

$$\mathbb{E}[V_{i+1}(X_{i+1}) \mid X_i = x] = \sum_{r=1}^M \beta_{ir} \psi_r(x), \quad (22.2)$$

for some basis functions $\psi_r : \mathbb{R}^d \rightarrow \mathbb{R}$ and constants $\beta_{ir}, r = 1, \dots, M$. Using the notation C_i for the continuation value at time i , we may equivalently write [Equation \(22.2\)](#) as

$$C_i(x) = \beta_i^\top \psi(x), \quad (22.3)$$

with

$$\beta_i^\top = (\beta_{i1}, \dots, \beta_{iM}), \quad \psi(x) = (\psi_1(x), \dots, \psi_M(x))^\top.$$

One could use different basis functions at different exercise dates, but to simplify notation, we suppress any dependence of ψ on i .

Assuming a relation of the form [Equation \(22.2\)](#) holds, the vector β_i is given by

$$\beta_i = (\mathbb{E}[\psi(X_i)\psi(X_i)^\top])^{-1} \mathbb{E}[\psi(X_i)V_{i+1}(X_{i+1})] \equiv B_\psi^{-1} B_{\psi V}. \quad (22.4)$$

Here, B_ψ is the indicated $M \times M$ matrix (assumed nonsingular) and $B_{\psi V}$ the indicated vector of length M . The variables (X_i, X_{i+1}) inside the expectations have a joint distribution corresponding to the state of the underlying Markov chain at dates i and $i + 1$.

The coefficients β_{ir} could be estimated from observations of pairs $(X_{ij}, V_{i+1}(X_{i+1,j}))$, $j = 1, \dots, b$, each consisting of the state at time i and the corresponding option value at time $i + 1$. Consider, in particular, independent paths $(X_{1j}, \dots, X_{mj})$, $j = 1, \dots, b$, and suppose for a moment that the values $V_{i+1}(X_{i+1,j})$ are known. The least-squares estimate of β_i is then given by

$$\hat{\beta}_i = \hat{B}_\psi^{-1} \hat{B}_{\psi V},$$

where $\hat{B}_\psi$ and $\hat{B}_{\psi V}$ are the sample counterparts of B_ψ and $B_{\psi V}$. More explicitly, $\hat{B}_\psi$ is the $M \times M$ matrix with qr entry

$$\frac{1}{b} \sum_{j=1}^b \psi_q(X_{ij}) \psi_r(X_{ij})$$

and $\hat{B}_{\psi V}$ is the M -vector with r th entry

$$\frac{1}{b} \sum_{k=1}^b \psi_r(X_{ik}) V_{i+1}(X_{i+1,k}). \quad (22.5)$$

All of these quantities can be calculated from function values at pairs of consecutive nodes $(X_{ij}, X_{i+1,j})$, $j = 1, \dots, b$. In practice, V_{i+1} is unknown and must be replaced by estimated values $\hat{V}_{i+1}$ at downstream nodes. The estimate $\hat{\beta}_i$ then defines an estimate

$$\hat{C}_i(x) = \hat{\beta}_i^\top \psi(x), \quad (22.6)$$

of the continuation value at an arbitrary point x in the state space $\mathbb{R}^d$. This, in turn, defines a procedure for estimating the option value.

Algorithm 1 Regression-Based Pricing Algorithm

- 1: Simulate b independent paths $\{X_{1j}, \dots, X_{mj}\}$, $j = 1, \dots, b$, of the Markov chain.
- 2: At terminal nodes, set $\hat{V}_{mj} = h_m(X_{mj})$ for $j = 1, \dots, b$.
- 3: **for** $i = m - 1$ **down to** 1 **do**
- 4: Given estimated values $\hat{V}_{i+1,j}$, $j = 1, \dots, b$, use regression to calculate $\hat{\beta}_i = \hat{B}_\psi^{-1} \hat{B}_{\psi V}$.
- 5: Compute estimated continuation values $\hat{C}_i(X_{ij})$ using [Equation \(22.6\)](#).
- 6: **for** $j = 1, \dots, b$ **do**
- 7: Set value based on immediate exercise versus estimated continuation:

$$\hat{V}_{ij} = \max \{h_i(X_{ij}), \hat{C}_i(X_{ij})\}. \quad (22.7)$$

- 8: **end for**
 - 9: **end for**
 - 10: Set $\hat{V}_0 = (\hat{V}_{11} + \dots + \hat{V}_{1b})/b$.
-

Definition 22.2: American Contingent Claim

An *American contingent claim* $X^a = (X, \mathcal{T}_{[0,T]})$ expiring at time T consists of a sequence of payoffs $(X_t)_{t=0}^T$, where X_t is an $\mathcal{F}_t$ -measurable random variable for $t = 0, 1, \dots, T$, and of the set $\mathcal{T}_{[0,T]}$ of admissible exercise policies.

Proposition 22.1: Arbitrage Price of American Put

The arbitrage price P_t^a of an American put option equals

$$P_t^a = \max_{\tau \in \mathcal{T}_{[t,T]}} \mathbb{E}_{\mathbb{P}^*} \left(\hat{r}^{-(\tau-t)} (K - S_\tau)^+ \mid \mathcal{F}_t \right), \quad \forall t \leq T. \quad (22.8)$$

Moreover, for any $t \leq T$ the stopping time τ_t^* that realizes the maximum in [Equation \(22.8\)](#) is given by the expression (by convention $\min \emptyset = T$)

$$\tau_t^* = \min \left\{ u \in \{t, t+1, \dots, T\} \mid (K - S_u)^+ \geq P_u^a \right\}. \quad (22.9)$$

Proposition 22.2: Arbitrage Price of American Claim in CRR

For every $t \leq T$, the arbitrage price $\pi(X^a)$ of an arbitrary American claim X^a in the CRR model equals

$$\pi_t(X^a) = \max_{\tau \in \mathcal{T}_{[t, T]}} \mathbb{E}_{\mathbb{P}^*} \left(\hat{r}^{-(\tau-t)} X_\tau \mid \mathcal{F}_t \right). \quad (22.10)$$

The price process $\pi(X^a)$ can be determined using the following recurrence relation, for every $t \leq T - 1$,

$$\pi_t(X^a) = \max \left\{ X_t, \mathbb{E}_{\mathbb{P}^*} \left(\hat{r}^{-1} \pi_{t+1}(X^a) \mid \mathcal{F}_t \right) \right\} \quad (22.11)$$

subject to the terminal condition $\pi_T(X^a) = X_T$. In the case of a path-independent American claim X^a with the reward function h we have that, for every $t \leq T - 1$,

$$\pi_t(X^a) = \max \left\{ h(S_t, t), \hat{r}^{-1} \left(p_* \pi_{t+1}^u(X^a) + (1 - p_*) \pi_{t+1}^d(X^a) \right) \right\}, \quad (22.12)$$

where, for a generic stock price value S_t at time t , we write $\pi_{t+1}^u(X^a)$ and $\pi_{t+1}^d(X^a)$ to denote the values of the price process $\pi(X^a)$ at time $t + 1$ in the nodes that correspond to the upward and downward movements of the stock price during the time-period $[t, t + 1]$, that is, for the values uS_t and dS_t of the stock price at time $t + 1$, respectively.

23 Basket Options

Appendices

A Existence of Lebesgue Measure

B Linear Algebra and Functional Analysis

C Multivariate Differentiation

D Probability and Statistics

E Python Primer